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### Mechanically locking actuator

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#### Abstract

A lockable telescopic actuator arranged to be in a locked state or in an unlocked state, the lockable telescopic actuator including: a body defining a cylindrical cavity; a piston mounted to slide in the cylindrical cavity; and a locking mechanism including a first locking segment arranged to move from a locked position to an unlocked position. In the locked position the first locking segment is engaged with the piston to prevent the piston from moving. In the unlocked position the first locking segment is disengaged from the piston to allow the piston to move. A first shape memory alloy, SMA, biasing element configured to be extended or compressed such that when the first SMA biasing element is compressed the first locking element is biased into the locked position and when the first SMA bias element is extended the first locking element is biased into the unlocked position.

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## Background/Summary

### BACKGROUND OF THE INVENTION

(1) Telescopic actuators are commonly used in airspace or other mechanical applications and generally comprise a cylinder or a body defining a cylindrical cavity and a piston or rod, which is slidably mounted to the body. There is often a need for telescopic actuators to be selectively lockable, i.e., to be configured such that in a first, locked, state the piston is prevented from moving relative to the body, while in a second, unlocked, state the piston is allowed to move relative to the body. It is generally desired to ensure that a lockable actuator will not prematurely transition from the locked state to the unlocked state or vice versa.

(2) In known lockable actuators, to ensure that there will be no premature transition between the locked and the unlocked state, the systems often rely on a resistive force, i.e., a frictional force, that needs to be overcome for locking and unlocking by spring and/or hydraulic systems. This can cause excessive wear of parts and may lead to higher maintenance requirements, impacting cost, reliability and potentially safety.

(3) There is thus a need for an improved lockable actuator that can reliably prevent an unwanted transition from a locked state to an unlocked state.

### SUMMARY OF THE INVENTION

(4) According to a first aspect of the invention there is provided a lockable telescopic actuator arranged to be in a locked state or in an unlocked state, the lockable telescopic actuator comprising: a body defining a cylindrical cavity; a piston mounted to slide telescopically in the cylindrical cavity; and a locking mechanism comprising a first locking segment, the first locking segment arranged to move from a locked position to an unlocked position, wherein in the locked position the first locking segment is engaged with the piston to prevent the piston from moving relative to the body and in the unlocked position the first locking segment is disengaged from the piston to allow the piston to move relative to the body; and a first shape memory alloy, SMA, biasing element configured to be selectively extended or compressed, the first SMA biasing element arranged such that when the first SMA biasing element is selectively compressed the first locking element is biased into the locked position and when the first SMA bias element is selectively extended the first locking element is biased into the unlocked position.

(5) Advantageously, the invention reduces wear in components of the actuator, as it does not have to rely on friction between the locking segments and the piston to maintain the locked or unlocked state. Furthermore, the invention requires fewer moving parts than known purely mechanical locking mechanisms, and thus enjoy improved reliability, reduced weight, and easier maintenance when compared with known lockable telescopic actuators.

(6) According to a second aspect of the invention there is provided a lockable telescopic actuator arranged to be in a locked state or in an unlocked state, the lockable telescopic actuator comprising: a body defining a cylindrical cavity; a piston mounted to slide telescopically in the cylindrical cavity; and a locking mechanism comprising a first locking segment, the first locking segment arranged to move from a locked position to an unlocked position, wherein in the locked position the first locking segment is engaged with the piston to prevent the piston from moving relative to the body and in the unlocked position the first locking segment is disengaged from the piston to allow the piston to move relative to the body; and a first shape memory alloy, SMA, biasing element configured to be selectively extended or compressed, the first SMA biasing element arranged such that when the first SMA biasing element is selectively extended the first locking element is biased into the locked position and when the first SMA bias element is selectively compressed the first locking element is biased into the unlocked position.

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# Description

## BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Embodiments of the invention will now be described, by way of non-limiting examples only, with reference to the following accompanying figures, in which:
- (2) FIG. **1a** is a schematic cross sectional diagram of an internally locking telescopic actuator with shape memory alloy in an unlocked state.
- (3) FIG. **1b** is a schematic cross sectional diagram of an internally locking telescopic actuator with shape memory alloy in a locked state.
- (4) FIG. **2a** is a partial perspective schematic diagram of an externally locking telescopic actuator in an unlocked state.
- (5) FIG. **2b** is a partial perspective schematic diagram of an externally locking telescopic actuator in a locked state.
- (6) FIG. **3a** is a cross sectional diagram of an externally locking telescopic actuator with shape memory alloy in an unlocked state.
- (7) FIG. **3b** is a cross sectional diagram of an externally locking telescopic actuator with shape memory alloy in a locked state.

## DETAILED DESCRIPTION OF EMBODIMENTS OF THE INVENTION

- (8) The present invention relates to a lockable telescopic actuator, comprising a cylinder or a body defining a cylindrical cavity and a piston or a rod, the piston or rod mounted to slide telescopically in the cylindrical cavity, the lockable telescopic actuator further comprising a locking mechanism arranged to be in a locked state or in an unlocked state. When the lockable telescopic actuator is in a locked state the piston is prevented from moving relative to the body. When the lockable telescopic actuator is in the unlocked state the piston is allowed to move relative to the body. A lockable telescopic actuator according to the invention can be incorporated in an landing gear for an aircraft, or in other components of an aircraft.
- (9) Shape Memory Alloys, SMAs, are a type of materials with the ability to return to a predefined shape after experiencing plastic deformation. SMAs revert back to their predefined shape by heating, cooling, or some other external stimulus as long as the experienced deformation is within a recoverable range. The process of deformation and shape recovery can be repeated many times.
- (10) SMAs undergo temperature driven phase transformation. Temperature change may be achieved by applying a current to the SMA, which due to the resistivity of the SMA can result in heating of the SMA. In some embodiments of the invention, the SMA is nickel-titanium alloy or Nitinol that has an appropriate resistivity for such applications. As the SMA changes temperature due to heating or cooling the crystal structure of the SMA changes which results in dimensional changes that can range between 1% and 8%. This can result in the SMA element extending or compressing depending on its temperature.
- (11) The SMA biasing elements discussed below may be electrically coupled to a current source (not shown in the following figures) to receive current. When current flows through any of the SMA biasing elements mentioned below, due to the resistivity of the SMA, the SMA may be heated to a first predefined temperature, at which point the SMA may have a first size or length. When current does not flow through the SMA, the SMA cools, at which point the SMA may have a second size or length. As such each SMA discussed later can be extended or compressed selectively, by heating or cooling the SMA.
- (12) FIG. **1a** schematically illustrates a cross sectional view of a telescopic actuator according to an embodiment of the invention, illustrated in the unlocked state. FIG. **1b** illustrates the same actuator of FIG. **1a** in the locked state.
- (13) The telescopic actuator **10** comprises a piston **16** mounted to slide telescopically in the cylindrical cavity of the body **15**, commonly known as a sliding piston. The piston **16** may

comprise dynamic seals **17** on a surface to reduce the possibility of leaking of gas or fluids contained in the telescopic actuator.

(14) The telescopic actuator **10** comprises a locking mechanism. The locking mechanism comprises a first locking segment **14a**, a second locking segment **14b**, and a first shape memory alloy, SMA, biasing element **12**. In the embodiment described in FIGS. **1a** and **1b**, the first locking segment **14a** and the second locking segment **14b** are shaped as rocking claws that may form jaws, although in other embodiments they may have different shapes as will be shown later in the present description. The first locking segment **14a** and the second locking segment **14b** are respectively attached on one side through hinges in the cylindrical cavity of the body **15** and arranged to pivot about each respective hinge.

(15) The piston **16** comprises on its surface at least one recess or slot **19**. In the embodiment illustrated in FIGS. **1a** and **1b** the first locking segment **14a** and the second locking segment **14b** are arranged to pivot about their respective hinges such that they can at least partially enter the recess **19** or be entirely out of the recess **19**. In a first state when the first locking segment **14a** and/or the second locking segment **14b** are at least partially inserted in the recess **19** the piston is prevented from moving relative to the body. In this first state the first locking segment **14a** and/or the second locking segment **14b** are in a locked position and the lockable telescopic actuator is in a locked state. In a second state when the first locking segment **14a** and the second locking segment **14b** are entirely out of the recess **19** the piston is allowed to move relative to the body. In the second state the first locking segment **14a** and/or the second locking segment **14b** are in an unlocked position and the lockable telescopic actuator is in an unlocked state.

(16) The telescopic actuator illustrated in FIGS. **1a** and **1b** also comprises a first SMA biasing element configured to be selectively extended or compressed. The first SMA is positioned between the first locking element **14a** and the second locking element **14b**.

(17) When the first SMA is selectively compressed the first locking element **14a** and the second locking element **14b** are biased such that they are in the locked position. In other words when the first SMA is selectively compressed first SMA applies to the first locking element **14a** and the second locking element **14b** a force such that the first locking element **14a** and the second locking element **14b** pivot about their respective hinges towards the recess until the first locking element **14a** and the second locking element **14b** are at least partially inserted in the recess **19** and the telescopic actuator is, as shown in FIG. **1b**.

(18) When the first SMA biasing element **12** is selectively extended the first locking element **14a** and the second locking element **14b** are biased such that they are in the unlocked position. In other words, when the first SMA is selectively extended, first SMA applies to the first locking element **14a** and the second locking element **14b** a force such that the first locking element **14a** and the second locking element **14b** pivot about their respective hinges away from the recess, until the first locking element **14a** and the second locking element **14b** are out of the recess **19** and the telescopic actuator is as shown in FIG. **1a**.

(19) The telescopic actuator illustrated in FIGS. **1a** and **1b** may also comprise a second SMA biasing element **13a** and a third SMA biasing element **13b** configured to be selectively extended or compressed. The second SMA biasing element **13a** is located in the cylindrical cavity and positioned between the first locking element **14a** and the body. The third SMA biasing element **13b** is located in the cylindrical cavity and positioned between the second locking element **14b** and the body.

(20) When the first SMA biasing element **12** is selectively extended the second SMA biasing element **13a** is selectively compressed. Thus, when the first SMA biasing element **12** applies a force to the first locking segment **14a** to move the first locking segment **14a** to the unlocked position, the second SMA biasing element **13a** also applies a force to the first locking segment **14a** to move the first locking segment **14a** to the unlocked position. Similarly, when the first SMA biasing element **12** is selectively compressed the second SMA biasing element **13a** is selectively

extended. Thus, when the first SMA biasing element **12** applies a force to the first locking segment **14a** to move the first locking segment **14a** to the locked position, the second SMA biasing element **13a** also applies a force to the first locking segment **14a** to move the first locking segment **14a** to the locked position. Advantageously, relying on two SMA biasing springs to move the first locking segment allows higher locking loads to be achieved and improved reliability.

(21) Similarly, when the first SMA biasing element **12** is selectively extended the third SMA biasing element **13b** is selectively compressed. Thus, when the first SMA biasing element **12** applies a force to the second locking segment **14b** to move the second locking segment **14b** to the unlocked position, the third SMA biasing element **13b** also applies a force to the second locking segment **14b** to move the second locking segment **14b** to the unlocked position. Similarly, when the first SMA biasing element **12** is selectively compressed the third SMA biasing element **13b** is selectively extended. Thus, when the first SMA biasing element **12** applies a force to the second locking segment **14b** to move the second locking segment **14b** to the locked position, the third SMA biasing element **13b** also applies a force to the second locking segment **14b** to move the second locking segment **14b** to the locked position. As in the case of the first SMA biasing element **12** and the second biasing element **13a**, relying on two SMA biasing springs to move the second locking segment **14b** allows higher locking loads to be achieved and improved reliability.

(22) Although the embodiment illustrated in FIGS. **1a** and **1b** comprises the first, second and third SMA biasing elements **12**, **13a** and **13b**, other embodiments may comprise only the first SMA biasing element **12**. Yet, even other embodiments may lack the first SMA biasing element **12** but have the second and/or the third SMA biasing elements **13a** and **13b**. Furthermore, other embodiments may have only the first locking segment **14a** and lack a further locking element, in which case the first SMA biasing element **12** may be between the first locking segment **14a** and the body **15**.

(23) In some embodiments of the invention, the telescopic actuator may comprise multiple locking mechanisms, each comprising respective rocking claws disposed radially with respect to the longitudinal axis of the piston.

(24) In operation, the telescopic actuator may initially be in an unlocked state as described above when a controller (not shown) enables a current source (not shown) to provide a first current to the first SMA biasing element **12** such that the first SMA biasing element **12** is maintained at a predefined first temperature, while providing a second current to the second and third SMA biasing elements **13a**, **13b** such that the second and third SMA biasing elements **13a**, **13b** are maintained at a predefined second temperature. At the predefined first temperature the first SMA biasing element **12** is extended and at the predefined second temperature the second and third SMA biasing elements **13a**, **13b** are compressed, thus biasing the first and second locking segments **14a**, **14b** in the unlocking position and allowing the piston **26** to move relative to the body **25**.

(25) When there is a need to lock the telescopic actuator the controller enables the current source to provide a predefined third current to the first SMA biasing element **12**, wherein the third current is less than the first current, such that the first SMA biasing element **12** is cooled to a predefined third temperature, while providing a fourth current to the second and third SMA biasing elements **13a**, **13b**, wherein the fourth current is greater than the second current such that the second and third SMA biasing elements **13a**, **13b** are heated to a predefined fourth temperature. The cooling of the first SMA biasing element **12** causes the first SMA biasing element **12** to compress, while the heating of the second and third SMA biasing elements **13a**, **13b** cause the second and third SMA biasing elements **13a**, **13b** to extend, biasing the first and second locking segments **14a**, **14b** in the locking position and preventing the piston **26** to move relative to the body **25**.

(26) When there is a need to unlock the telescopic actuator, the controller enables the current source to provide to the first SMA biasing element **12** the first current, thus extending the first SMA biasing element **12**, and to provide the second and third SMA biasing elements **13a**, **13b** the second current, thus compressing the second and third SMA biasing elements **13a**, **13b**. Extending

the first SMA biasing element **12** and compressing the second and third SMA biasing elements **13a**, **13b** biases the first and second locking segments **14a**, **14b** back to the unlocked position, allowing the piston **26** to move relative to the body **25**.

(27) The piston **16** illustrated in FIGS. **1a** and **1b** may comprise an indicator switch **18** in the recess **19**. The indicator switch **18** is configured to be activated when in contact with the first and/or second locking segments **14a**, **14b** and deactivated when the first and second locking segments are out of the recess **19**. In other embodiments, the indicator switch **18** may be configured to be deactivated when in contact with the first and/or second locking segments **14a**, **14b** and activated when the first and second locking segments are out of the recess **19**. The indicator switch **18** may be part of a feedback mechanism (not shown) that comprises the current source, thus providing a single centralised system that verifies that the heating or cooling of the SMA has reached the desired levels based on the output of the indicator switch **18** that provides feedback of whether the telescopic actuator is in the locked state or in the unlocked state.

(28) Advantageously, as each of the SMA biasing elements can separately bias the locking segments between the locked and the unlocked position, the actuator benefits from improved redundancy and reliability. In other words, even if the first SMA biasing element **12** fails, the second and third SMA biasing element **13a** and **13b** will still be able to move the first and second locking segments **14a**, **14b** to engage or disengage with the piston **16**, and vice versa.

(29) Although in the telescopic actuator illustrated in FIGS. **1a** and **1b** the cooling of the first SMA biasing element **12** causes the first SMA biasing element **12** to compress, while the heating of the second and third SMA biasing elements **13a**, **13b** causes the second and third SMA biasing elements **13a**, **13b** to extend, in other embodiments the SMA biasing elements may be arranged such that the heating of the first SMA biasing element **12** would cause the first SMA biasing element **12** to compress, while the cooling of the second and third SMA biasing elements **13a**, **13b** would cause the second and third SMA biasing elements **13a**, **13b** to extend, or in other words the opposite behaviour to the telescopic actuator illustrated in FIGS. **1a** and **1b**. Yet, in other embodiments, the first SMA biasing element **12** and the second and third SMA biasing elements **13a**, **13b** may comprise different SMA, such that the cooling of all of the first SMA biasing element **12** and the second and third SMA biasing elements **13a**, **13b** causes the first SMA biasing element **12** to compress and the second and third SMA biasing elements **13a**, **13b** to extend, or alternatively, the heating of all of the first SMA biasing element **12** and the second and third SMA biasing elements **13a**, **13b** causes the first SMA biasing element **12** to compress and the second and third SMA biasing elements **13a**, **13b** to extend. The skilled person will readily understand that based on the arrangement of the SMA biasing elements different combinations of SMAs can be employed to bias the telescopic actuator to the locked or unlocked position for a variety of temperatures.

(30) FIGS. **2a** and **2b** are perspective schematic diagrams of a telescopic actuator **20** according to another embodiment of the invention, wherein the locking mechanism of the lockable telescopic actuator is not located in the cylindrical cavity but is mounted externally on the body. FIGS. **3a** and **3b** are cross-sectional representations of the embodiment presented in FIGS. **2a** and **2b** accordingly. FIGS. **2a** and **3a** illustrate the externally locking telescopic actuator in the unlocked state, while FIGS. **2b** and **3b** illustrate the externally locking telescopic actuator in the locked state.

(31) The telescopic actuator **20** comprises a piston **26** mounted to slide telescopically in the cylindrical cavity of the body **25**, commonly known as a sliding piston. The piston **26** may comprise dynamic seals **27** (not shown) to reduce the possibility of leaking of gas or fluids contained in the telescopic actuator.

(32) The telescopic actuator **20** comprises a locking mechanism mounted on a housing **21**. The housing **21** is attached on the external surface of the body **25** with a pin **27**, although in other embodiments the housing **21** may be attached on the body in other manners. In the embodiment illustrated in FIGS. **2a-3b**, the housing **21** surrounds the body **25**, although in other embodiments the housing **21** may be in contact with only a small part of the external surface of the body **25**. The

locking mechanism comprises a first locking segment **24**, a first SMA biasing element **22** and a second SMA biasing element **23**.

(33) As can be seen in FIGS. **2a** and **2b** the telescopic actuator may have multiple locking mechanisms mounted on the housing **21** radially with respect to the longitudinal axis of the piston **26**, such that they surround the piston **26**, with each locking mechanism comprising a respective first locking segment, a first SMA biasing element and a second SMA biasing element.

(34) The first locking segment **24** is slidably engaged to the housing **21** through the first SMA biasing element **22** and the second SMA biasing element **23**. The first SMA biasing element **22** is arranged between the housing **21** and the first locking segment **24** such that when the first SMA biasing element **22** is selectively extended the first locking element **24** is moved away from the piston **26** while when the first SMA biasing element **22** is selectively compressed the first locking element **24** is moved towards the piston **26**. The second SMA biasing element **23** is arranged between the housing **21** and the first locking segment **24** such that when the second SMA biasing element **23** is selectively compressed the first locking element **24** is moved away from the piston **26** while when the second SMA biasing element **23** is selectively extended the first locking element **24** is moved towards the piston **26**.

(35) The piston **26** comprises on its surface at least one recess or slot **29**. In the embodiment illustrated in FIGS. **2a-3b** the first locking segment **24** is arranged such that it can move from being at least partially inserted in the recess **29** to be entirely out of the recess **29**. In a first state, when the first locking segment **24** is at least partially inserted in the recess **29** the piston is prevented from moving relative to the body **25**. In this first state, the first locking segment **24** is in a locked position and the lockable telescopic actuator is in a locked state. In a second state, when the first locking segment **24** is entirely out of the recess **29**, the piston **26** is allowed to move relative to the body **25**. In the second state the first locking segment **24** is in an unlocked position and the lockable telescopic actuator is in an unlocked state.

(36) In the embodiment described in FIGS. **2a-3b**, the first locking segment **24** is shaped as a cuboid, although in other embodiments the first locking segment **24** may be shaped as a pin or have other appropriate shapes such that it conforms to the shape of the recess **29**.

(37) In operation, the telescopic actuator may initially be in the unlocked state as described above when a controller (not shown) enables a current source (not shown) to provide a first current to the first SMA biasing element **22** such that the first SMA biasing element **22** is maintained at a respective predefined first temperature, while providing a second current to the second SMA biasing elements **23** such that the second SMA biasing elements **23** is maintained at a respective predefined second temperature. At the predefined first temperature the first SMA biasing element **22** is extended and at the predefined second temperature the second biasing elements **23** is compressed, thus biasing the first locking segment **24** in the unlocking position and allowing the piston **26** to move relative to the body **25**.

(38) When there is a need to lock the telescopic actuator, the controller enables the current source to provide a respective predefined third current to the first SMA biasing element **22**, wherein the respective third current is less than the respective first current, such that the first SMA biasing element **22** is cooled at a respective predefined third temperature, while providing a respective predefined fourth current to the second SMA biasing element **23**, wherein the respective fourth current is greater than the respective second current such that the second SMA biasing elements **23** is heated at a respective predefined fourth temperature. The cooling of the first SMA biasing element **22** causes the first SMA biasing element **22** to compress, while the heating of the second SMA biasing elements **23** causes the second SMA biasing element **23** to extend, biasing the first locking segments **24** in the locking position and preventing the piston **26** to move relative to the body **25**.

(39) When there is a need to unlock the telescopic actuator, the controller enables the current source to provide the respective predefined first current to the first SMA biasing element **22**, thus

extending the first SMA biasing element **22**, and to provide the respective predefined second current to the second SMA biasing elements **23**, thus compressing the second SMA biasing element **23**. Extending the first SMA biasing element **22** and compressing the second SMA biasing element **23** biases the first locking segment **24** back to the unlocked position, allowing the piston **26** to move relative to the body **25**.

(40) Advantageously as the first SMA biasing element **22** and the second biasing element **23** separately can bias the first locking segment **24** between the locked and the unlocked position, the actuator benefits from improved redundancy and reliability. In other words, even if the first biasing element **22** fails, the second biasing element **23** will still be able to move the first locking segments **24** to engage or disengage with the piston **26**, and vice versa.

(41) The first locking segment **24** may comprise an indication switch **28** on a surface facing the recess **29**. The indication switch **28** is configured to be activated when the first locking segment enters the recess **29** and deactivated when the first locking segment is out of the recess **29** or vice versa. The indication switch **28** may be part of a feedback mechanism (not shown) that comprises the current source, thus providing a single centralised system that verifies that the heating or cooling of the SMA has reached the desired levels based on the output of the indication switch **28** that provides feedback of whether the telescopic actuator is in the locked state or in the unlocked state.

(42) Although in the embodiment described above the current source changes the amplitude of the current supplied to the SMA biasing elements, in other embodiments, the current source may be configured to provide current pulses of fixed amplitude, and the cooling or heating of the SMA biasing elements may be adjusted by the current source changing the frequency of the current sources.

(43) Advantageously in the above locking mechanisms, locking and unlocking the actuator is controlled by electrical means thus requiring fewer moving parts than known purely mechanical locking mechanisms, and thus enjoy improved reliability, reduced weight, and easier maintenance when compared with known lockable telescopic actuators, which are all significant advantages in aerospace applications.

(44) Furthermore, in embodiments of the invention comprising multiple SMA biasing elements, for example as illustrated in the figures, actuation of the locking segments can be based solely on any one of the SMA biasing elements. The system thus benefits from redundancy and improved reliability.

(45) Although the invention has been described above with reference to one or more preferred embodiments, it will be appreciated that various changes or modifications may be made without departing from the scope of the invention as defined in the appended claims. The word “comprising” can mean “including” or “consisting of” and therefore does not exclude the presence of elements or steps other than those listed in any claim or the specification as a whole. The mere fact that certain measures are recited in mutually different dependent claims does not indicate that a combination of these measures cannot be used to advantage.

## Claims

1. A lockable telescopic actuator arranged to be in a locked state or in an unlocked state, the lockable telescopic actuator comprising: a body defining a cylindrical cavity; a piston mounted to slide telescopically in the cylindrical cavity; and a locking mechanism comprising a first locking segment, the first locking segment arranged to be movable between a locked position and an unlocked position, wherein in the locked position the first locking segment is engaged with the piston to prevent the piston from moving relative to the body and in the unlocked position the first locking segment is disengaged from the piston to allow the piston to move relative to the body; and a first shape memory alloy, SMA, biasing element configured to be selectively extended or

compressed, the first SMA biasing element arranged such that when the first SMA biasing element is selectively compressed, the first locking segment is biased into the locked position and when the first SMA bias element is selectively extended, the first locking element is biased into the unlocked position, wherein the locking mechanism is located in the cylindrical cavity and further comprises a second locking segment, the second locking segment arranged to be movable between a locked position and an unlocked position, wherein in the locked position the second locking segment is engaged with the piston to prevent the piston from moving relative to the body and in the unlocked position the second locking segment is disengaged from the piston to allow the piston to move relative to the body; and wherein the first SMA biasing element is located between the first and second locking segments and arranged such that when the first SMA biasing element is selectively compressed, the second locking element is biased into the locked position and when the first SMA bias element is selectively extended, the second locking element is biased into the unlocked position, comprising a second SMA biasing element arranged between the body and the first locking segment and a third SMA biasing element arranged between the body and the second locking segment, the second and third SMA biasing elements configured to be selectively extended or compressed and arranged to: compress when the first SMA biasing element extends; and extend when the first SMA biasing element compresses.

2. A lockable telescopic actuator according to claim 1, wherein the piston includes a recess formed therein and wherein in the locked position the first locking segment is at least partially inserted in the recess; and in the unlocked position the first locking segment is out of the recess.

3. A lockable telescopic actuator according to claim 1, wherein the piston and/or the first locking segment comprises an indication switch that is: activated when the first locking segment is engaged with the piston; and deactivated when the first locking segment is disengaged with the piston.

4. A lockable telescopic actuator according to claim 1, wherein the first SMA biasing element is arranged to extend when at a first predefined temperature and to compress when at a second predefined temperature.

5. A lockable telescopic actuator according to claim 1, wherein the first SMA biasing element is electrically coupled to a current source.

6. A landing gear for an aircraft, comprising a lockable telescopic actuator according to claim 1.

7. A lockable telescopic actuator arranged to be in a locked state or in an unlocked state, the lockable telescopic actuator comprising: a body defining a cylindrical cavity; a piston mounted to slide telescopically in the cylindrical cavity; and a locking mechanism comprising a first locking segment, the first locking segment arranged to be movable between a locked position and an unlocked position, wherein in the locked position the first locking segment is engaged with the piston to prevent the piston from moving relative to the body and in the unlocked position the first locking segment is disengaged from the piston to allow the piston to move relative to the body; and a first shape memory alloy, SMA, biasing element configured to be selectively extended or compressed, the first SMA biasing element arranged such that when the first SMA biasing element is selectively extended, the first locking segment is biased into the locked position and when the first SMA bias element is selectively compressed, the first locking element is biased into the unlocked position, wherein the locking mechanism is mounted on a housing attached on an external surface of the body, the first SMA biasing element being located between the housing and the first locking segment, wherein the first locking segment is slidably engaged with the housing, and wherein the lockable telescopic actuator further comprised a second SMA biasing element located between the housing and the first locking segment, the second SMA biasing element configured to be selectively extended or compressed, the second SMA biasing element arranged: to compress when the first SMA biasing element extends; and to extend when the first SMA biasing element compresses.

8. A lockable telescopic actuator according to claim 7, wherein the piston includes a recess formed therein and wherein in the locked position the first locking segment is at least partially inserted in

the recess; and in the unlocked position the first locking segment is out of the recess.

9. A lockable telescopic actuator according to claim 7, wherein the piston and/or the first locking segment comprises an indication switch that is: activated when the first locking segment is engaged with the piston; and deactivated when the first locking segment is disengaged with the piston.

10. A lockable telescopic actuator according to claim 7, wherein the first SMA biasing element is arranged to extend when at a first predefined temperature and to compress when at a second predefined temperature.

11. A lockable telescopic actuator according to claim 7, wherein the first SMA biasing element is electrically coupled to a current source.

12. A landing gear for an aircraft, comprising a lockable telescopic actuator according to claim 7.

13. A lockable telescopic actuator arranged to be in a locked state or in an unlocked state, the lockable telescopic actuator comprising: a body defining a cylindrical cavity; a piston mounted to slide telescopically in the cylindrical cavity; and a locking mechanism comprising: a first locking segment, the first locking segment arranged to be movable between a locked position and an unlocked position, wherein in the locked position the first locking segment is engaged with the piston to prevent the piston from moving relative to the body and in the unlocked position the first locking segment is disengaged from the piston to allow the piston to move relative to the body; a first shape memory alloy, SMA, biasing element configured to be selectively extended or compressed, the first SMA biasing element arranged such that when the first SMA biasing element is selectively extended, the first locking segment is biased into the locked position and when the first SMA bias element is selectively compressed, the first locking element is biased into the unlocked position; wherein the locking mechanism is located in the cylindrical cavity and comprises a second locking segment, the second locking segment arranged to be movable between a locked position and an unlocked position, wherein in the locked position the second locking segment is engaged with the piston to prevent the piston from moving relative to the body and in the unlocked position the second locking segment is disengaged from the piston to allow the piston to move relative to the body; and a second SMA biasing element configured to be selectively extended or compressed, the second SMA biasing element arranged such that when the first SMA biasing element is selectively extended, the second SMA biasing element is selectively extended and the second locking element is biased into the locked position and when the first SMA bias element is selectively compressed, the second SMA biasing element is selectively compressed and the second locking element is biased into the unlocked position wherein the first SMA biasing element is between the body and the first locking segment and the second SMA biasing element is between the body and the second locking segment.

14. A lockable telescopic actuator according to claim 13, wherein the piston includes a recess formed therein and wherein in the locked position the first locking segment is at least partially inserted in the recess; and in the unlocked position the first locking segment is out of the recess.

15. A lockable telescopic actuator according to claim 13, wherein the piston and/or the first locking segment comprises an indication switch that is: activated when the first locking segment is engaged with the piston; and deactivated when the first locking segment is disengaged with the piston.

16. A lockable telescopic actuator according to claim 13, wherein the first SMA biasing element is arranged to extend when at a first predefined temperature and to compress when at a second predefined temperature.

17. A lockable telescopic actuator according to claim 13, wherein the first SMA biasing element is electrically coupled to a current source.

18. A landing gear for an aircraft, comprising a lockable telescopic actuator according to claim 13.

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